OFFICIAL COURSE PROBLEM SOLUTIONS • ZERO SKIPPED STEPS

Topic 3: Financial Statements (Exercises 1 to 6)

Source Document: `T3_Financial Statements_EXERCISES.pdf` (UC3M Business Management) • Solved with exhaustive mathematical and accounting steps, exam-day strategies, and balanced statements.

OFFICIAL MATERIAL `T3_Financial Statements_EXERCISES.pdf`

`Topic 3 Financial Management (I).pdf`

`Exercise 1 • Foundational Accounting Principles`

Accrual vs Cash Basis, Linear Depreciation & Inventory Sensitivity

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> *A specific company has started activity on the 1st of January of the present year. To this date (31st of March), it has the following goods that were acquired on 31st of December of the previous year:*
> *• Factory building: 1.000 m.u. (Duration 25 years)*
> *• Office building: 1.500 m.u. (Duration 50 years)*
> *• Machinery: 300 m.u. (Duration 10 years)*
> *• Transport equipment: 100 m.u. (Duration 5 years)*
> *Information corresponding to the 1st quarter is as follows: personnel salaries of the quarter 450 m.u. (75 m.u. pending after 31/3), materials purchases and used in production 700 m.u. (125 m.u. pending after 31/3), energy consumption estimated in 230 m.u. (nothing paid till date), sales invoiced to customers 1.930 m.u. (315 m.u. pending after 31/3). In manufacturing all sold units, the entire purchased materials have been consumed. Ignore corporation taxes.*
> ***a)*** *Calculate at 31/3: income, expenses, collections/cash inflows, payments/cash outflows, investments and depreciation/amortization of the quarter (assume linear depreciation of the goods along their lifetime).*
> ***b)*** *How will the response change if only half of the purchased materials had been consumed?*

EXAM DAY STRATEGY: ACCRUAL VS CASH & QUARTERLY PRO-RATA ALIGNMENT

- **Horizon & Pro-Rata Time:** 1st Quarter ($t = 3$ months $= \frac{1}{4}$ year $= 0.25$). Goods bought on 31/12 of the previous year $\implies$ exactly one quarter of depreciation has accrued on 31/3. Straight-line formula: $D_{\text{quarter}} = \frac{\text{Acquisition Cost}}{4 \times n}$.
- **Accrual Principle (Devengo) vs Cash Flow (Tesorería):**
  - **Revenues (Ingresos):** Recognized upon commercial delivery and invoice generation ($1,930$ mu), independent of payment terms.
  - **Collections (Cobros):** Physical cash collected $= 1,930 - 315 = \mathbf{1,615}$ mu. The pending 315 mu becomes an asset (Accounts Receivable) on the Balance Sheet.
  - **Expenses (Gastos):** Consumed productive factors and non-cash asset wear: Salaries (450) + Materials consumed (700) + Energy consumed (230) + Depreciation (30) $= \mathbf{1,410}$ mu.
  - **Payments (Pagos):** Actual treasury disbursements $= (450 - 75) + (700 - 125) + 0 = 375 + 575 = \mathbf{950}$ mu.
- **Exam Trap 1 – Depreciation Cash Flow:** Depreciation (30 mu) is an accounting charge for physical/technological wear and tear. It entails **zero cash payment (**0 mu**)**. Never add depreciation to cash outflows!
- **Exam Trap 2 – Capital Investments (Capex):** The $2,900$ mu of fixed assets were purchased on 31/12 prior year. During the 1st quarter itself, no additional fixed assets were acquired $\implies$ Q1 Capex $= \mathbf{0}$ mu.
- **Exam Trap 3 – Inventory Sensitivity (Part b):** If only 50% of materials are consumed (350 mu), the remaining 350 mu is stored as an **Inventory Asset**. Operating expenses drop by 350 mu, but **supplier payments remain strictly** 575 mu because payments depend on purchases invoiced, not physical consumption!

Phase 1: Quarterly Linear Depreciation Computation

The time elapsed is the 1st Quarter ($t = 3$ months $= \frac{1}{4}$ year). The straight-line quarterly depreciation allocates asset cost evenly over its lifespan:

$$D_{\text{quarter}} = \frac{\text{Annual Depreciation}}{4} = \frac{\text{Acquisition Cost}}{4 \times n}$$

- **Factory building:** $\frac{1,000}{4 \times 25} = \frac{1,000}{100} = \mathbf{10.0}$ m.u.
- **Office building:** $\frac{1,500}{4 \times 50} = \frac{1,500}{200} = \mathbf{7.5}$ m.u.
- **Machinery:** $\frac{300}{4 \times 10} = \frac{300}{40} = \mathbf{7.5}$ m.u.
- **Transport equipment:** $\frac{100}{4 \times 5} = \frac{100}{20} = \mathbf{5.0}$ m.u.
- **Total Depreciation for the Quarter:** $10.0 + 7.5 + 7.5 + 5.0 = \mathbf{30.0}$ m.u.

Phase 2: Resolution of Part (a) — 100% Consumed

1. **Income / Revenues (Ingresos):** Under the accrual principle, revenue is recognized upon commercial invoicing regardless of collection date:

$$\text{Income} = \text{Total Sales Invoiced} = \mathbf{1,930.0}\ \text{m.u.}$$

2. **Expenses (Gastos):** Value of all factor inputs consumed in operations during the quarter:

$$\begin{aligned} \text{Salaries Expense} &= 450.0\ \text{m.u.} \\ \text{Materials Consumed} &= 700.0\ \text{m.u.} \quad (100\%\ \text{consumed in production}) \\ \text{Energy Consumed} &= 230.0\ \text{m.u.} \\ \text{Depreciation} &= 30.0\ \text{m.u.} \\ \hline \text{Total Expenses} &= 450.0 + 700.0 + 230.0 + 30.0 = \mathbf{1,410.0}\ \text{m.u.} \end{aligned}$$

(Quarterly Net Profit $= \text{Income} - \text{Expenses} = 1,930.0 - 1,410.0 = \mathbf{520.0}$ m.u.*).*

3. **Collections / Cash Inflows (Cobros):** Actual liquid treasury received into the bank till:

$$\text{Collections} = \text{Total Invoiced} - \text{Pending at } 31/3 = 1,930.0 - 315.0 = \mathbf{1,615.0}\ \text{m.u.}$$

(The uncollected 315.0 m.u. *is a Current Asset: Accounts Receivable).*

4. **Payments / Cash Outflows (Pagos):** Physical disbursements from company treasury:

$$\begin{aligned} \text{Salaries Paid} &= 450.0 - 75.0 = 375.0\ \text{m.u.} \quad (\text{with 75 mu Salaries Payable}) \\ \text{Materials Paid} &= 700.0 - 125.0 = 575.0\ \text{m.u.} \quad (\text{with 125 mu Accounts Payable}) \\ \text{Energy Paid} &= 230.0 - 230.0 = 0.0\ \text{m.u.} \quad (\text{entirely pending liability}) \\ \text{Depreciation Paid} &= 0.0\ \text{m.u.} \quad (\text{non-cash accounting charge}) \\ \hline \text{Total Payments} &= 375.0 + 575.0 + 0.0 + 0.0 = \mathbf{950.0}\ \text{m.u.} \end{aligned}$$

(Net Cash Generated in Quarter $= \text{Collections} - \text{Payments} = 1,615.0 - 950.0 = +\mathbf{665.0}$ m.u.*).*

5. **Investments (Inversiones):** Initial productive capacity was capitalized on 31/12 of the previous year ($1,000 + 1,500 + 300 + 100 = 2,900.0$ m.u.). During Q1, no new fixed capital acquisitions occurred:

$$\text{Q1 Fixed Capital Investments} = \mathbf{0.0}\ \text{m.u.}$$

Alternative reading: the statement does not say whether "investments of the quarter" means new purchases or the capital invested in the goods listed. If it is read as the productive capital capitalised on 31/12 and in use during the quarter, the answer is $1,000 + 1,500 + 300 + 100 = 2,900$ m.u.

6. **Depreciation / Amortization:** Straight-line quarterly asset wear $=$ **30.0** m.u.

Phase 3: Resolution of Part (b) — Inventory Sensitivity (50% Consumed)

If only half of the purchased materials are consumed in production ($700.0/2 = 350.0$ m.u.), the remaining 350.0 m.u. remains untouched in the warehouse as a balance sheet asset:

- **Operating Expenses:** Materials expense drops from 700 mu to 350 mu:

$$\text{New Expenses} = 450.0 + \mathbf{350.0} + 230.0 + 30.0 = \mathbf{1,060.0}\ \text{m.u.} \quad (\Delta\text{Expenses} = -350.0\ \text{m.u.})$$

$$\text{New Net Income} = 1,930.0 - 1,060.0 = \mathbf{870.0}\ \text{m.u.} \quad (\text{increased by } +350.0\ \text{m.u.})$$

- **Payments / Cash Outflows: Completely unchanged at** 950.0 m.u.**!** Supplier payments are governed by raw materials purchased and credit terms ($700 - 125 = 575$ m.u.), regardless of physical shop-floor consumption.
- **Incomes and Collections:** Unchanged at $1,930.0$ m.u. and $1,615.0$ m.u. respectively.
- **Balance Sheet Asset Impact:** Warehouse Ending Inventory increases by $+350.0$ m.u. under Current Assets.

Financial Magnitude	Part (a): 100% Consumed	Part (b): 50% Consumed	Variation Δ
Income (Revenues)	1,930.0 m.u.	1,930.0 m.u.	0
Operating Expenses	1,410.0 m.u.	1,060.0 m.u.	-350.0 m.u.
Net Income (Profit)	**520.0 m.u.**	**870.0 m.u.**	**+350.0 m.u.**
Collections (Cash Inflows)	1,615.0 m.u.	1,615.0 m.u.	0
Payments (Cash Outflows)	950.0 m.u.	950.0 m.u.	0
Net Treasury Cash Flow	**+665.0 m.u.**	**+665.0 m.u.**	**0**
Ending Inventory Asset	0.0 m.u.	350.0 m.u.	+350.0 m.u.
Quarterly Depreciation	30.0 m.u.	30.0 m.u.	0

PEDAGOGICAL CONCLUSION & GOLDEN RULE

Unconsumed purchased materials represent a **Balance Sheet Current Asset (Inventory)**, not an Income Statement expense. While Net Income increases by 350 m.u., treasury cash flow remains identical. **Profit $\neq$ Cash Flow!**

Exercise 2 • Opening Balance Sheet & Mixed Contributions

JUGA S.A. Opening Balance Sheet with In-Kind Contributions

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> *The JUGA S.A. Firm is started with a shareholders contribution of 60.000 Euros. Of those 30.000 are paid in and deposited in a Bank account and the remaining are corresponding to the transfer of a patent for a value of 30.000 Euros. With the cash of the Bank account, raw materials have been purchased for a total amount of 10.500 Euros and machinery for 15.000 Euros. Please detail the balance sheet at the operations start.*

EXAM DAY STRATEGY: IN-KIND CONTRIBUTIONS & CASH LEDGER TRACKING

- **Initial Equity Structure:** Subscribed Share Capital $= €60,000$.
  - Monetary cash contributed to bank account: $+€30,000$.
  - In-kind non-monetary contribution: Industrial Patent valued at $+€30,000 \implies$ **Intangible Non-Current Asset**.
- **Asset Reallocation Funded by Bank Cash:** The problem explicitly states that purchases are made *with the cash of the bank account*:
  - Raw materials purchased: $+€10,500$ (Current Asset: Inventories).
  - Machinery purchased: $+€15,000$ (Non-Current Tangible Asset: PP&E).
  - Total cash disbursed: $10,500 + 15,000 = €25,500$.
  - **Ending Bank Balance:** $30,000 - 25,500 = €\mathbf{4,500}$.
- **Exam Trap 1 – Raw Materials Treatment:** Since the firm is at the *operations start* (no manufacturing has occurred and no sales have been made), raw materials are **100% an Inventory Asset** on the Balance Sheet, NOT an expense on an income statement.
- **Exam Trap 2 – Bank Balance Adjustment:** Do not leave the bank balance at €30,000 while also adding machinery and raw materials. That would cause assets to total €85,500 against €60,000 of equity, violating the fundamental accounting identity!
- **Exam Trap 3 – Patent Classification:** An industrial patent is an *Intangible Fixed Asset (Inmovilizado Intangible)*, not an expense and not a current asset.
- **Exam-Day Checkpoint:** Verify that Total Assets $= 45,000\ (\text{NCA}) + 15,000\ (\text{CA}) = €\mathbf{60,000} \equiv$ Stockholders' Equity $= €\mathbf{60,000}$, with Liabilities $= €\mathbf{0}$.

Phase 1: Step-by-Step Journaling & Cash Ledger

1. **Initial Capital Subscription:**

$$\text{Subscribed Share Capital} = \mathbf{60,000}\ €$$

Formed by:

- Bank account deposit (Monetary): $+30,000\ €$

- Industrial Patent contribution (In-kind Intangible Asset): $+30,000$ €

2. **Asset Acquisitions Funded from Bank Liquidity:**

$$\begin{aligned} \text{Raw Materials Purchased (Inventory)} &= +10,500 \text{ €} \quad \text{(Current Asset)} \\ \text{Machinery Purchased (PP\&E)} &= +15,000 \text{ €} \quad \text{(Non-Current Tangible Asset)} \\ \hline \text{Total Cash Outflows} &= 10,500 + 15,000 = \mathbf{25,500} \text{ €} \end{aligned}$$

3. **Ending Treasury Cash Balance at Bank:**

$$\text{Bank Account Balance} = 30,000 - 25,500 = \mathbf{4,500} \text{ €}$$

Phase 2: The Formal Opening Balance Sheet

ASSETS (Activo)	**Amount (€)**	**LIABILITIES & EQUITY (Pasivo y PN)**	**Amount (€)**
NON-CURRENT ASSETS (Activo No Corriente)	**45,000**	**STOCKHOLDERS' EQUITY (Patrimonio Neto)**	**60,000**
• Intangible Assets: Industrial Patent	30,000	• Subscribed Share Capital	60,000
• Tangible Fixed Assets: Machinery	15,000		
CURRENT ASSETS (Activo Corriente)	**15,000**	**NON-CURRENT LIABILITIES (Pasivo No Corriente)**	**0**
• Inventories: Raw Materials	10,500	• Long-term bank debt	0
• Cash & Cash Equivalents (Bank account)	4,500	**CURRENT LIABILITIES (Pasivo Corriente)**	**0**
		• Accounts payable (Suppliers)	0
TOTAL ASSETS	**60,000**	**TOTAL LIABILITIES & EQUITY**	**60,000**

STRUCTURAL & SOLVENCY VERDICT

$$\text{Assets } (\mathbf{60,000} \text{ €}) \equiv \text{Liabilities } (\mathbf{0} \text{ €}) + \text{Equity } (\mathbf{60,000} \text{ €})$$

The company starts with zero debt leverage ($\text{Debt}/\text{Equity} = 0$), financed 100% by equity capital. Working capital is $\text{WC} = \text{CA} - \text{CL} = 15,000 - 0 = +\mathbf{15,000}$ €, providing absolute short-term liquidity safety.

Exercise 3 • Balance Sheet Classification & Working Capital

Libby's Company Balance Sheet Classification & Solvency Audit

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 2

> *Prepare the BALANCE SHEET of Libby's Company using the following figures (in euros): Short-term loan debts 15.500 | Long-term loan debts 3.000 | Money on its bank account 3.000 | Cash 500 | Paid-in capital 12.000 | Accounts receivable 3.500 | Accounts payable 1.500 | Equipment 13.000 | Raw materials (purchased this year but not used yet) 12.500 | Furniture 1.500 | Retained earnings from the year 2.000.*

EXAM DAY STRATEGY: STANDARD ACCOUNT CLASSIFICATION & DOUBLE WC VERIFICATION

- **5-Block Balance Sheet Architecture:** Always group accounts into standard financial blocks:
  1. **Non-Current Assets (Activo No Corriente):** Productive durable capital (> 1 year) $\implies$ Equipment (€13,000) + Furniture (€1,500) = $\mathbf{€14,500}$.
  2. **Current Assets (Activo Corriente):** Operating cycle liquid items (≤ 1 year) $\implies$ Raw materials inventory (€12,500) + Accounts receivable (€3,500) + Bank (€3,000) + Cash (€500) = $\mathbf{€19,500}$.
  3. **Stockholders' Equity (Patrimonio Neto):** Owner funding & internal retained profits $\implies$ Paid-in capital (€12,000) + Retained earnings (€2,000) = $\mathbf{€14,000}$.
  4. **Non-Current Liabilities (Pasivo No Corriente):** External obligations with maturity > 1 year $\implies$ Long-term loan debts = $\mathbf{€3,000}$.
  5. **Current Liabilities (Pasivo Corriente):** Short-term obligations due within 1 year $\implies$ Short-term loan debts (€15,500) + Accounts payable (€1,500) = $\mathbf{€17,000}$.
- **Dual Verification of Working Capital (Fondo de Maniobra):**

$$\text{Method 1 (Operational): } \mathbf{WC} = \text{Current Assets} - \text{Current Liabilities} = 19,500 - 17,000 = +\mathbf{2,500}\ €$$

$$\text{Method 2 (Financing): } \mathbf{WC} = (\text{Equity} + \text{Non-Current Liabilities}) - \text{Non-Current Assets} = (14,000 + 3,000) - 14,500 = +\mathbf{2,500}\ €$$

  (Both methods MUST yield identical values. If they differ, an account was misclassified).
- **Exam Trap 1 – Retained Earnings Placement:** "Retained earnings from the year" represents accumulated profits belonging to owners; it is part of **Stockholders' Equity**, never a liability!
- **Exam Trap 2 – Receivables vs Payables:** "Accounts receivable" (Deudores comerciales/Clientes) is an Asset (money owed to Libby's); "Accounts payable" (Acreedores/Proveedores) is a Current Liability (money Libby's owes to suppliers).
- **Exam Trap 3 – Loan Debt Horizon:** Short-term loan debt (€15,500) must be clearly separated from long-term loan debt (€3,000). Grouping them together forfeits solvency points.

Phase 1: Standard Account Categorization Matrix

Assigning each ledger line item to its statutory destination:

- **Non-Current Assets:** Equipment (€13,000) + Furniture (€1,500) = **€14,500**.
- **Current Assets:** Raw materials inventory (€12,500) + Accounts receivable (€3,500) + Bank (€3,000) + Cash (€500) = **€19,500**.
- **Stockholders' Equity:** Paid-in capital (€12,000) + Retained earnings (€2,000) = **€14,000**.
- **Non-Current Liabilities:** Long-term loan debts = **€3,000**.
- **Current Liabilities:** Short-term loan debts (€15,500) + Accounts payable (€1,500) = **€17,000**.

Phase 2: Formal Balanced Balance Sheet

ASSETS (Activo)	**Amount (€)**	**LIABILITIES & EQUITY (Pasivo y PN)**	**Amount (€)**
NON-CURRENT ASSETS (Activo No Corriente)	**14,500**	**STOCKHOLDERS' EQUITY (Patrimonio Neto)**	**14,000**
• Tangible Fixed Assets: Equipment	13,000	• Paid-in Share Capital	12,000
• Tangible Fixed Assets: Furniture	1,500	• Retained Earnings from the year	2,000
CURRENT ASSETS (Activo Corriente)	**19,500**	**NON-CURRENT LIABILITIES (Pasivo No Corriente)**	**3,000**
• Inventories (Raw materials not used yet)	12,500	• Long-term loan debts	3,000
• Accounts Receivable (Customers)	3,500	**CURRENT LIABILITIES (Pasivo Corriente)**	**17,000**
• Cash & Cash Equivalents: Bank account	3,000	• Short-term loan debts	15,500
• Cash & Cash Equivalents: Cash on hand	500	• Accounts payable (Trade suppliers)	1,500
TOTAL ASSETS	**34,000**	**TOTAL LIABILITIES & EQUITY**	**34,000**

SOLVENCY, LIQUIDITY & LEVERAGE RATIOS

1. Working Capital: $\text{WC} = \text{CA} - \text{CL} = 19,500 - 17,000 = +\mathbf{2,500}$ € (Positive safety cushion: permanent resources finance all non-current assets plus 12.8% of current assets).

2. Current Ratio (Short-Term Solvency): $\text{CR} = \frac{\text{CA}}{\text{CL}} = \frac{19,500}{17,000} = \mathbf{1.147}$ (Adequate coverage > 1.0, but tightly bound to inventory liquidation).

3. Acid-Test (Quick Ratio): $\text{QR} = \frac{\text{Cash} + \text{Receivables}}{\text{CL}} = \frac{3,500 + 3,500}{17,000} = \frac{7,000}{17,000} = \mathbf{0.412}$ (Highlights that 64.1% of current assets are tied up in raw materials).

4. Debt Structure & Leverage: $\text{Total Debt} = 3,000 + 17,000 = 20,000$ €. Debt-to-Equity $= \frac{20,000}{14,000} = \mathbf{1.429}$. Crucially, $\frac{15,500 + 1,500}{20,000} = \frac{17,000}{20,000} = \mathbf{85\%}$ of total indebtedness (short-term loans plus trade payables) matures within a year, requiring proactive refinancing; the short-term loans alone are $\frac{15,500}{20,000} = 77.5\%$.

Exercise 4 • Multi-Step Income Statement

Hawkins Partners Income Statement & Inventory Adjustments

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Prepare the Income Statement for Hawkins Partners using the following figures and knowing that corporate tax applicable rate is 30% (figures in euros):
1. Initial inventory: 600 | 2. Purchase of goods (sold 100%): 84.000 | 3. Rental costs: 1.000 | 4. Property sold to another company (income): 10.500 | 5. Interests received: 15.200 | 6. Interests from long-term debt: 5.500 | 7. Wages and salaries: 12.700 | 8. Water, gas and electricity: 1.200 | 9. Sales: 130.800 | 10. Inventory at end of period: 500.

EXAM DAY STRATEGY: INVENTORY VARIATION, EXTRAORDINARY GAINS & MULTI-STEP P&L

- **Standard Multi-Step P&L Structure:**

$$\text{Sales Revenue} \overset{-\text{COGS}}{\longrightarrow} \text{Gross Profit} \overset{-\text{Op. Expenses}}{\longrightarrow} \text{EBIT} \overset{+\text{Non-Op}\pm\text{Financial}}{\longrightarrow} \text{EBT} \overset{-\text{Taxes}}{\longrightarrow} \text{Net Income}$$

- **Cost of Goods Sold (COGS) Identity:**

$$\text{COGS} = \text{Beginning Inventory} + \text{Purchases} - \text{Ending Inventory}$$

$$\text{COGS} = 600 + 84,000 - 500 = \mathbf{84,100}\ €$$

Economic intuition: Ending inventory dropped by $100\ €$ ($600 - 500$). This inventory depletion was sold to customers, so it must be added to purchases ($84,000 + 100 = 84,100$).

- **Exam Trap 1 – Extraordinary Property Sale:** "Property sold to another company (€10,500)" is a capital gain from the disposal of a non-current asset, **NOT regular commercial turnover**! It must NOT be mixed into operating sales, but placed under **Non-Operating / Extraordinary Income**.
- **Exam Trap 2 – Financial Interests Separation:** "Interests received (€15,200)" is financial income; "Interests from long-term debt (€5,500)" is financial expense. Net Financial Result $= +15,200 - 5,500 = +\mathbf{9,700}\ €$.
- **Exam Trap 3 – Corporate Tax Base:** Apply the 30% tax rate to **EBT** (€52,000), NEVER to EBIT (€31,800). Corporate income tax covers all profits: operating, extraordinary, and financial.

Phase 1: Step-by-Step Arithmetic Deductions

1. **Cost of Goods Sold (COGS):**

$$\text{COGS} = \text{Initial Inventory} + \text{Purchases} - \text{Ending Inventory} = 600 + 84,000 - 500 = \mathbf{84,100}\ €$$

2. **Gross Profit (Margen Bruto):**

$$\text{Gross Profit} = \text{Sales Revenue} - \text{COGS} = 130,800 - 84,100 = \mathbf{46,700}\ €$$

3. **Operating Expenses (Gastos de Explotación):**

$$\begin{aligned} \text{Wages and salaries} &= 12,700\ € \\ \text{Rental costs} &= 1,000\ € \\ \text{Water, gas and electricity (Utilities)} &= 1,200\ € \\ \hline \text{Total Operating Expenses} &= 12,700 + 1,000 + 1,200 = \mathbf{14,900}\ € \end{aligned}$$

4. **Operating Income / EBIT (Resultado de Explotación):**

$$\text{EBIT} = \text{Gross Profit} - \text{Operating Expenses} = 46,700 - 14,900 = \mathbf{31,800}\ €$$

5. **Non-Operating & Financial Items (Resultados Financieros y Extraordinarios):**

$$\begin{aligned} \text{Gain on property sold (Non-operating)} &= +10,500\ € \\ \text{Interests received (Financial income)} &= +15,200\ € \\ \text{Interests on debt (Financial expense)} &= -5,500\ € \\ \hline \text{Net Non-Operating \& Financial} &= +10,500 + 15,200 - 5,500 = +\mathbf{20,200}\ € \end{aligned}$$

6. **Earnings Before Taxes / EBT (Resultado Antes de Impuestos):**

$$\text{EBT} = \text{EBIT} + \text{Net Non-Operating \& Financial} = 31,800 + 20,200 = \mathbf{52,000}\ €$$

7. **Corporate Income Tax (30%) and Net Profit:**

$$\text{Corporate Tax} = 30\% \times \text{EBT} = 0.30 \times 52,000 = \mathbf{15,600}\ €$$

$$\text{Net Income (Net Profit)} = \text{EBT} - \text{Corporate Tax} = 52,000 - 15,600 = \mathbf{36,400}\ €$$

Phase 2: Formal Multi-Step Income Statement

HAWKINS PARTNERS – INCOME STATEMENT	Detail (€)	Amount (€)
Sales Revenue (Turnover)		130,800
(-) Cost of Goods Sold (COGS)		(84,100)
• Initial inventory	600	
• Purchases of merchandise	84,000	
• (-) Ending inventory	(500)	
= GROSS PROFIT (Margen Bruto)		**46,700**
(-) Operating Expenses		(14,900)
• Personnel (Wages and salaries)	(12,700)	
• Rental costs	(1,000)	
• Utilities (Water, gas, electricity)	(1,200)	
= OPERATING INCOME / EBIT (Resultado de Explotación)		**31,800**
(+) Extraordinary Gain: Property sold to another company		+10,500
(+) Financial Income: Interests received		+15,200
(-) Financial Expenses: Interests on long-term debt		(5,500)
= EARNINGS BEFORE TAXES / EBT (Resultado Antes de Impuestos)		**52,000**
(-) Corporate Income Tax (30%)		(15,600)
NET INCOME / NET PROFIT (Resultado del Ejercicio)		**36,400**

ECONOMIC MARGIN INSIGHTS

Gross Margin: $\frac{\text{Gross Profit}}{\text{Sales}} = \frac{46,700}{130,800} = \mathbf{35.70\%}$

Operating Margin (ROS): $\frac{\text{EBIT}}{\text{Sales}} = \frac{31,800}{130,800} = \mathbf{24.31\%}$

Net Margin: $\frac{\text{Net Income}}{\text{Sales}} = \frac{36,400}{130,800} = \mathbf{27.83\%}$ (Boosted significantly by extraordinary property gain and financial investments).

Corporate Income Statement, Tiered Discounts, Taxes & Dividends

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 2

Prepare the Income Statement of a company with the following information from year 2017:

- *Gross sales: 2.002 million €. Trade policy gives 3% discount to Type A customers (annual purchases > 120 million €). Type A customers account for 40% of sales. Total sales = Gross sales - Discount.*
- *Purchases of raw materials: 962 million €. Assume no change in stocks.*
- *Workforce: 7.800 employees, average salary 30.000 €/year, Social Security spending 30% of salary.*
- *Other operating expenses: 457 million €.*
- *In recent years invested 1.200 million € in fixed assets (lifespan 10 years, linear depreciation, no residual value).*
- *International loan obtained in 2015 of 225 million € at 6% interest along the whole year, principal repayable in 2020.*
- *Corporate tax rate: 30%. Agreed dividend: 50%.*

EXAM DAY STRATEGY: DISCOUNT ELIGIBILITY, SOCIAL SECURITY & DIVIDEND FLOW

- **Commercial Discount Eligibility:**

$$\text{Type A Customer Purchases} = 40\% \times 2,002.00 \text{ M€} = \mathbf{800.80} \text{ M€}$$

The 120 M€ threshold is a per-customer criterion that defines who is Type A; the statement already gives their aggregate share (40% of sales), so no test of 800.80 against 120 is needed. The 3% discount applies to that aggregate:

$$\text{Discount} = 3\% \times 800.80 = \mathbf{24.024} \text{ M€} \implies \text{Net Sales} = 2,002.00 - 24.024 = \mathbf{1,977.976} \text{ M€}$$

- **Workforce Labor Cost (Gross Salary + Employer Social Security):**

$$\text{Base Wages} = 7,800 \times 30,000 \text{ €} = 234.00 \text{ M€}$$

$$\text{Employer Social Security} = 30\% \times 234.00 = 70.20 \text{ M€} \implies \text{Total Labor Expense} = \mathbf{304.20} \text{ M€}$$

- **Exam Trap 1 – Discount Scope:** Never apply the 3% discount to the full 2,002 M€! The prompt states only Type A customers (40%) qualify. Applying the discount to total sales deducts 60.06 M€ instead of 24.024 M€, distorting all downstream figures.
- **Exam Trap 2 – Principal Repayment vs Interest:** The loan principal (225 M€) matures in 2020. **Principal repayments are Balance Sheet cash/debt movements, NEVER Income Statement expenses!** Only the annual interest ($6\% \times 225 = 13.50$ M€) is recorded in the P&L.
- **Exam Trap 3 – Dividend Payout Base:** Apply the 50% dividend rate to **Net Income (profit after corporate taxes)**, never to EBIT or EBT. Dividends represent profit distribution to shareholders after all statutory obligations are fulfilled.

Phase 1: Step-by-Step Numerical Deductions

1. **Gross Sales, Commercial Discount & Net Sales:**

$$\text{Type A Volume} = 0.40 \times 2,002.00 = 800.80 \text{ M€} \quad \text{(the 120 M€ threshold is per customer and only defines the Type A class)}$$

$$\text{Commercial Discount} = 0.03 \times 800.80 = \mathbf{24.024} \text{ M€}$$

$$\text{Net Sales Turnover} = 2,002.00 - 24.024 = \mathbf{1,977.976}\ \text{M€}$$

2. **Cost of Goods Sold (COGS):** Because raw material stocks did not change ($\Delta\text{Stock} = 0$):

$$\text{COGS} = \text{Purchases of Raw Materials} = \mathbf{962.000}\ \text{M€}$$

$$\text{Gross Profit} = \text{Net Sales} - \text{COGS} = 1,977.976 - 962.000 = \mathbf{1,015.976}\ \text{M€}$$

3. **Operating Expenses (Gastos de Explotación):**

$$\begin{aligned}
\text{Base Salaries (7,800 workers} \times 30,000\ €) &= 234.000\ \text{M€} \\
\text{Employer Social Security (30\% on salaries)} &= +70.200\ \text{M€} \\
\text{Total Personnel Expense} &= \mathbf{304.200}\ \text{M€} \\
\text{Other Operating Expenses} &= \mathbf{457.000}\ \text{M€} \\
\text{Linear Depreciation } \left(\frac{1,200.00}{10\ \text{yrs}}\right) &= \mathbf{120.000}\ \text{M€} \\
\hline
\text{Total Operating Expenses} &= 304.200 + 457.000 + 120.000 = \mathbf{881.200}\ \text{M€}
\end{aligned}$$

4. **Operating Income / EBIT (Resultado de Explotación):**

$$\text{EBIT} = \text{Gross Profit} - \text{Operating Expenses} = 1,015.976 - 881.200 = \mathbf{134.776}\ \text{M€}$$

5. **Financial Expenses (Intereses de la Deuda):** Outstanding loan balance in 2017 is 225 M€ at 6% per annum:

$$\text{Financial Interest Expense} = 0.06 \times 225.00 = \mathbf{13.500}\ \text{M€}$$

6. **Earnings Before Taxes / EBT (Resultado Antes de Impuestos):**

$$\text{EBT} = \text{EBIT} - \text{Interest Expense} = 134.776 - 13.500 = \mathbf{121.276}\ \text{M€}$$

7. **Corporate Income Taxes (30%) & Net Income:**

$$\text{Corporate Tax} = 30\% \times \text{EBT} = 0.30 \times 121.276 = \mathbf{36.3828}\ \text{M€}$$

$$\text{Net Income (Net Profit)} = 121.276 - 36.3828 = \mathbf{84.8932}\ \text{M€}$$

8. **Profit Appropriation (Dividends vs Retained Reserves):**

$$\text{Dividends to Shareholders (50\% Payout)} = 0.50 \times 84.8932 = \mathbf{42.4466}\ \text{M€}$$

$$\text{Retained Earnings to Equity Reserves (50\% Retention)} = 0.50 \times 84.8932 = \mathbf{42.4466}\ \text{M€}$$

Phase 2: Formal Corporate Income Statement (Year 2017)

INCOME STATEMENT – FISCAL YEAR 2017	Detail (M€)	Amount (M€)
Gross Sales Turnover	2,002.000	
(-) Commercial Discounts (3% on Type A customers)	(24.024)	
= **NET SALES REVENUE**		**1,977.976**
(-) Cost of Goods Sold (Raw materials purchases)		(962.000)
= **GROSS PROFIT**		**1,015.976**
(-) Operating Expenses		(881.200)
• Personnel Cost (Wages 234 M€ + 30% SS 70.2 M€)	(304.200)	
• Other Operating Expenses	(457.000)	
• Linear Depreciation of Fixed Assets (10% of 1,200 M€)	(120.000)	
= **OPERATING INCOME / EBIT**		**134.776**
(-) Financial Interest Expenses (6% of 225 M€ loan)		(13.500)
= **EARNINGS BEFORE TAXES / EBT**		**121.276**
(-) Corporate Income Tax (30%)		(36.383)
NET INCOME FOR THE YEAR		**84.893**
• Dividends to be distributed (50% payout)		42.447
• Retained Earnings transferred to Equity reserves (50%)		42.447

FINANCIAL RATIOS & PAYOUT SUMMARY

Operating Return on Sales (ROS): $\frac{\text{EBIT}}{\text{Net Sales}} = \frac{134.776}{1,977.976} = \mathbf{6.81\%}$

Net Profit Margin: $\frac{\text{Net Income}}{\text{Net Sales}} = \frac{84.893}{1,977.976} = \mathbf{4.29\%}$

Interest Coverage Ratio (TIE): $\frac{\text{EBIT}}{\text{Interest}} = \frac{134.776}{13.500} = \mathbf{9.98}\times$ (Strong solvency: operating profit covers debt interest nearly 10 times over).

Exercise 6 • Full Accounting Cycle & Uncalled Capital

Company X Full Accounting Cycle, Balance Sheet & Uncalled Capital

Original statement · T3_Financial Statements_EXERCISES.pdf, p. 3

> *Company X incorporated on 1/1/2017 with 100 m.u. shares fully paid in cash. Purchased on 1/1/2017 with cash: a) land/area for 20 m.u., b) equipment with life of 10 years (straight-line) for 20 m.u., c) raw materials for 20 m.u.*
> *Hires an employee for annual cost of 10 m.u. Employee processes half of raw material and sells it for 50 m.u. Sales are uniform every day, average collection period is 3 months. Corporate tax is 30% (paid on 31/12/2017). Strict self-financing strategy.*
> ***1.** Prepare the Balance Sheet as on 31/12/2017 with intermediate calculations.*
> ***2.** Indicate and explain changes if shareholders had paid only 90% of the 100 m.u. subscribed.*

EXAM DAY STRATEGY: FULL ANNUAL CYCLE, TREASURY LEDGER & CONTRA-EQUITY

- **Opening Cash Position on 1/1/2017:**

$$\text{Cash}_{1/1} = \text{Share Capital} - (\text{Land} + \text{Equipment} + \text{Materials}) = 100.0 - (20.0 + 20.0 + 20.0) = \mathbf{40.0}\ \text{mu}$$

- **Operating Cycle & COGS:** Only half of raw materials are processed $\implies$ $\text{COGS} = 20.0/2 = \mathbf{10.0}$ mu. The remaining 10.0 mu remains in warehouse inventory as a **Current Asset**.
- **Exam Trap 1 – Depreciation of Land: Land is indestructible and never depreciates!** Only Equipment depreciates ($20.0/10 = \mathbf{2.0}$ mu). Depreciating land is an immediate exam penalty.
- **Exam Trap 2 – Accounts Receivable Delay:** Sales occur uniformly over 365 days with an average collection period of 3 months (0.25 year).

$$\text{Ending Receivables} = 50.0 \times \frac{3\ \text{months}}{12\ \text{months}} = \mathbf{12.5}\ \text{mu} \implies \text{Collections} = 50.0 - 12.5 = \mathbf{37.5}\ \text{mu}$$

  Never assume that all 50 mu of sales were collected in cash!
- **Exam Trap 3 – Corporate Tax Settlement:** The problem states that corporate tax ($30\% \times 28 = 8.4$ mu) is *paid on December 31st, 2017*. Thus, it is a **cash outflow** during 2017, leaving **zero tax liability** pending on the year-end balance sheet!
- **Exam Trap 4 – Question 2 (Uncalled Capital):** If shareholders only pay 90%, uncalled capital is 10 mu. Uncalled capital does NOT change revenues, expenses, or taxes. It decreases starting cash by 10 mu (ending cash becomes 49.1 mu) and is presented in Equity as a negative contra-equity line ($100 - 10 + 19.6 = 109.6$ mu).

Phase 1: Operational Cycle & P&L Statement (Question 1)

1. **Sales Revenue:** Sales $= \mathbf{50.0}$ m.u.

2. **Raw Materials Consumed (COGS):**

$$\text{COGS} = \frac{20.0}{2} = \mathbf{10.0}\ \text{m.u.} \quad (\text{Ending Inventory stored in warehouse} = 10.0\ \text{m.u.})$$

$$\text{Gross Profit} = 50.0 - 10.0 = \mathbf{40.0}\ \text{m.u.}$$

3. **Operating Expenses (Gastos de Explotación):**
   - Employee labor cost: 10.0 m.u.
   - Equipment depreciation ($20/10$ yrs): 2.0 m.u.

$$\text{Total Operating Expenses} = 10.0 + 2.0 = \mathbf{12.0}\ \text{m.u.}$$

4. **Operating Income (EBIT) & EBT:**

$$\text{EBIT} = 40.0 - 12.0 = \mathbf{28.0}\ \text{m.u.} \implies \text{EBT} = \mathbf{28.0}\ \text{m.u.}$$

5. **Corporate Income Tax (30% paid in cash on 31/12):**

$$\text{Corporate Tax} = 30\% \times 28.0 = \mathbf{8.4}\ \text{m.u.}$$

6. **Net Income & Self-Financing Appropriation:**

$$\text{Net Income} = 28.0 - 8.4 = \mathbf{19.6}\ \text{m.u.}$$

Under strict self-financing, $\text{Dividends} = 0$ m.u. and the full profit is retained into reserves:

$$\text{Retained Earnings (Addition to Equity)} = +\mathbf{19.6}\ \text{m.u.}$$

Phase 2: Working Capital & Treasury Cash Reconciliation

1. **Accounts Receivable (Trade Debtors):**

$$\text{Ending Accounts Receivable} = \text{Sales} \times \frac{3\ \text{months}}{12\ \text{months}} = 50.0 \times \frac{1}{4} = \mathbf{12.5}\ \text{m.u.}$$

$$\text{Cash Collections from Customers} = 50.0 - 12.5 = \mathbf{37.5}\ \text{m.u.}$$

2. **Treasury Cash Ledger Reconciliation on 31/12/2017:**

Cash Ledger Item	Amount (m.u.)
Cash balance on 1/1/2017 (100 equity − 60 capex & stock)	40.0
(+) Collections from customers	+37.5
(-) Employee salaries paid	(10.0)
(-) Corporate tax paid on 31/12	(8.4)
Ending Cash Balance on 31/12/2017	**59.1**

Phase 3: Balance Sheet as on 31/12/2017 (100% Paid — Question 1)

ASSETS (Activo)	Amount (m.u.)	LIABILITIES & EQUITY (Pasivo y PN)	Amount (m.u.)
NON-CURRENT ASSETS (Activo No Corriente)	**38.0**	**STOCKHOLDERS' EQUITY (Patrimonio Neto)**	**119.6**
• Land / Industrial Area (Non-depreciable)	20.0	• Subscribed & Paid-in Share Capital	100.0
• Productive Equipment (Historical cost)	20.0	• Retained Earnings (Year 2017 Net Profit)	19.6
• (-) Accumulated Depreciation: Equipment	(2.0)		
CURRENT ASSETS (Activo Corriente)	**81.6**	**NON-CURRENT LIABILITIES**	**0.0**
• Inventories: Unprocessed Raw Materials	10.0	• Long-term bank borrowing	0.0
• Accounts Receivable (Customers, 3-month term)	12.5	**CURRENT LIABILITIES**	**0.0**
• Cash & Cash Equivalents (Treasury account)	59.1	• Accounts payable / Unpaid taxes	0.0
TOTAL ASSETS	**119.6**	**TOTAL LIABILITIES & EQUITY**	**119.6**

Phase 4: Impact of Uncalled Capital (90% Paid — Question 2)

If shareholders contributed only 90% of subscribed shares in cash (90 m.u.), the remaining 10 m.u. represents **Uncalled Capital** (*Accionistas por desembolsos no exigidos*):

1. **Income Statement & Net Profit:** Completely unaffected (Net Income = **19.6** m.u.). Operations, billing, wages, depreciation, and taxes do not depend on the equity disbursement fraction.

2. **Cash & Current Assets:** Starting cash on 1/1/2017 was $90.0 - 60.0 = 30.0$ m.u. (exactly 10.0 m.u. lower).

$$\text{New Ending Cash} = 59.1 - 10.0 = \mathbf{49.1} \text{ m.u.}$$

$$\text{New Current Assets} = 10.0 + 12.5 + 49.1 = \mathbf{71.6} \text{ m.u.}$$

$$\text{New Total Assets} = 38.0 \text{ (NCA)} + 71.6 \text{ (CA)} = \mathbf{109.6} \text{ m.u.}$$

3. **Stockholders' Equity (Statutory Contra-Equity Presentation):** Under standard European / Spanish General Accounting Plan rules, uncalled capital appears as a negative contra-equity line directly beneath subscribed capital:

Equity Component (Question 2)	Amount (m.u.)
Subscribed Share Capital	100.0
(-) Shareholders for Uncalled Capital	(10.0)
(+) Retained Earnings (Year 2017)	+19.6
Total Stockholders' Equity	**109.6**

BALANCE CHECK WITH UNCALLED CAPITAL

$$\text{Total Assets } (\mathbf{109.6} \text{ m.u.}) \equiv \text{Total Liabilities \& Equity } (\mathbf{109.6} \text{ m.u.})$$

Alternative Accounting Presentation Note: Under US GAAP, uncalled capital is occasionally presented as a receivable asset. However, under standard Spanish PGC and IAS/IFRS, presenting it as a negative contra-equity line within Net Equity is the standard, ensuring that net equity accurately reflects resources physically contributed plus retained profit.

Study notes written from the official course material (T3_Financial Statements_EXERCISES.pdf; Topic 3 Financial Management (I).pdf). Online version: subjects/business-management/problemas/topic-3-financial-statements.html